

Shreyas Patange

+91 8956380952 patangeshreyas@gmail.com [LinkedIn](#) [GitHub](#) [LeetCode](#) [Portfolio](#)

Experience

Skylark Labs

July 2025 - Present

Software Development Engineer II

- Built a **multi-region distributed system** for **RTSP/WebRTC** video ingestion, egress, and processing using **TypeScript**, **Go**, **OpenCV**, **FFmpeg**, and **gRPC**, sustaining **10K+ concurrent live streams** with real-time analytics at **sub-400ms p95** end-to-end latency.
- Developed an **agentic AI platform** using **Mastra** and **TypeScript**, implementing hybrid retrieval (metadata filtering, semantic search, and structured querying), tool calling, contextual memory, and autonomous workflows, enabling natural-language analytics over **1M+** surveillance events and cutting manual review time by **60%**.
- Implemented **Envoy** and **Traefik** load balancing for **gRPC** and **HTTP** services using **Maglev consistent hashing**, distributing **10K+** live streams across processing nodes and enabling **A/B deployments** with **Kubernetes** and **Docker**.
- Designed and built a centralized **identity and access management** platform supporting **OAuth2**, **JWT**-based authentication, organization management, licensing, and fine-grained **RBAC**, serving multiple internal services and handling **100K+** auth and licensing requests monthly.
- Managed **multi-region AWS infrastructure** (**Kubernetes**, **ECS**, **Lambda**, **RDS**) behind **Cloudflare global load balancing**, with **Terraform**-based IaC, **CI/CD** pipelines, and full observability via **Prometheus**, **Grafana**, and **Loki**.
- Improved system reliability by integrating **AWS SQS/SNS** distributed queues and building the real-time alerting stack (WebSockets, push, email, SMS), delivering **100K+ alerts/day** with **under 2s** median delivery latency.

Heliverse Technologies

Dec 2023 - July 2025

Tech Lead

- Led a team of **5 engineers**; built an **AI-powered real-time voice agent** and **WhatsApp agent** using **LangChain** and **Twilio**, handling **100+ conversations/day** via **BullMQ** workflows and automating **70%** of inquiries end-to-end.
- Developed a **serverless microservice architecture** on **AWS Lambda** with **DynamoDB**, **TypeScript**, and **CloudFront**, deployed via the **Serverless Framework**.
- Optimized frontend performance using **Next.js SSG/SSR** and server actions for type-safe **RPC** calls.
- Architected an HR platform backend with **NestJS**, **Prisma**, and **PostgreSQL**, featuring shift management, biometric attendance, and payroll generation for **100+ employees**, with real-time notifications via **Firebase** and **WebSockets**.

Projects

GeoQuiz | [React](#), [Redux](#), [Supabase](#), [TypeScript](#), [WebSockets](#) [Demo](#) [GitHub](#)

- Developed a **Street View**-based location guessing game with **single-player** and **real-time multiplayer** lobbies using **TypeScript**, **React**, and the **Google Maps Street View API** along with **Supabase**.
- Implemented **low-latency multiplayer** logic using **Supabase WebSocket**-driven real-time database streams, enabling synchronized room state, competitive rounds, chat, reconnection logic, and failure handling.
- Designed deterministic scoring logic and synchronized multiplayer game state using **PostgreSQL** and Supabase real-time infrastructure for consistent gameplay across clients.

Gor Programming Language | [Go](#), [Gin](#), [React](#) [Demo](#) [GitHub](#)

- Built a **Turing-complete interpreted programming language** from scratch with a full **tokenizer**, **parser**, **AST** representation, **optimizer**, **evaluation engine**, and interactive **REPL**.
- Created an **online playground** for writing and running Gor code, visualizing ASTs, and interacting with the language through a **React + Gin** web interface.

Technical Skills

Languages: TypeScript/JavaScript, Golang, Python, C/C++, Rust, Zig, Bash, SQL

Frameworks & Technologies: React, Next.js, NestJS, FastAPI, Gin, Hono, Bun.js, GraphQL, gRPC, WebRTC, WebSocket, WebTransport, FFmpeg, NATS, Kafka, RabbitMQ, Temporal, LangChain

Infrastructure & DevOps: Docker, Kubernetes, Traefik, NGINX, Terraform, Helm, Linux, CI/CD (GitHub Actions)

Databases: PostgreSQL, CockroachDB, MySQL (Vitess), MongoDB, DynamoDB, ScyllaDB, Cassandra, Redis

Cloud Platforms: AWS (ECS, Lambda, RDS, S3, EC2), Cloudflare, GCP, Firebase, Supabase

Exposure: Java, C#, Spring Boot, .NET, Lua, WASM, ONNX, Onvif, NVIDIA Triton

Concepts: Distributed Systems, Consensus, System Design, Load Balancing, Microservices, Networking, Operating Systems, Databases, Algorithms and Data Structures, RAG, Agentic AI

Achievements

- Reached LeetCode **Knight** rank (top 5%) with a **2000+** contest rating and **1100+** problems solved.

Education

University of Mumbai (GPA: 9.30 / 10.00)

Bachelor of Engineering in Computer Science, with minor in Data Science

2020-2024

Thane, Maharashtra